

B-Learner: Quasi-Oracle Bounds on Heterogeneous Causal Effects Under Hidden Confounding

Causal Inference in the AI Era

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Today: two acts

Part 1 — The paper

- Partial identification and bounds
- Sensitivity analysis lineage: ATE $\rightarrow$ CATE
- Tan's MSM (Λ); $\Lambda = 2$ example
- Valid vs. sharp bounds; sharp CATE bounds
- Why CVaR = worst-case confounding
- Plug-in bias, pseudo-outcome, Neyman orthogonality
- Theorem 1: full bias bound
- Pointwise vs. average validity
- Empirical results

Part 2 — Referee mode

- Assumptions: tested vs. untested
- Limitations
- Comparison to classical methods
- Discussion

Paper

Oprescu et al. *B-Learner: Quasi-Oracle Bounds on Heterogeneous Causal Effects Under Hidden Confounding*. ICML 2023.

The problem: CATE under hidden confounding

Standard CATE meta-learners (T-, X-, DR-Learner) require:

$$(Y(1), Y(0)) \perp\!\!\!\perp T \mid X \quad (\text{unconfoundedness})$$

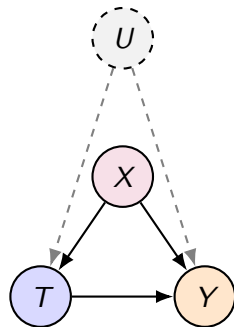
In practice there may be a hidden U affecting both T and Y . Unconfoundedness fails; any CATE estimate may be arbitrarily biased.

What can we do?

Instead of a point estimate $\hat{\tau}(x)$, learn a bound:

$$[\hat{\tau}^-(x), \hat{\tau}^+(x)]$$

that is *provably valid* under a specified limit on how much U can confound.



Hidden U : unconfoundedness fails.
CATE estimate $\hat{\tau}(x)$ has unknown bias.

Key concept: partial identification and the identified set

Point identification

We can recover $\tau(x)$ exactly from observational data under unconfoundedness. The estimand is a **single value**.

Partial identification

When unconfoundedness fails, $\tau(x)$ is **not unique**. Many DGPs are consistent with the data. Each implies a different $\tau(x)$.

The **identified set** is the set of all $\tau(x)$ values that some consistent DGP can produce.

A **bound** $[\tau^-(x), \tau^+(x)]$ characterizes this set.

What do we need?

- An assumption that *limits* (not eliminates) the hidden confounding
- Under that limit, the identified set becomes a bounded interval

Nuisance functions

Intermediate functions we estimate not for their own sake, but because we need them to estimate $\tau(x)$:

- Propensity score $e(x)$
- Outcome regression $\mu(x, a)$
- Quantile function $q(x, a)$ (new here)

Where does B-Learner come from? Methodological lineage

Building block 1: Dorn, Guo & Kallus (2021) — sharp ATE bounds

Derived **sharp** bounds on the average treatment effect under Tan's MSM.

Key tool: CVaR as the worst-case potential outcome distribution.

Provides the formula for $Y^+(x, a)$ that B-Learner builds on (our Result 1).

Building block 2: Kallus & Oprescu (2023) — doubly-robust meta-learning

Framework for doubly-robust, model-agnostic learning of **conditional** distributional effects.

Provides the Neyman-orthogonal pseudo-outcome construction and the second-stage regression machinery.

B-Learner (this paper)

Combines both: applies Kallus & Oprescu's efficient pseudo-outcome machinery to the Dorn et al. sharp ATE bound formula, **lifting ATE bounds to CATE bounds** with quasi-oracle rates, double robustness, and flexible learners.

Tan's Marginal Sensitivity Model (MSM)

Let $e(x) = P(T = 1 \mid X = x)$ be the **observed** propensity score.

Let $e(x, u) = P(T = 1 \mid X = x, U = u)$ be the **full** propensity (unobservable).

Assumption 1 (MSM): There exists $\Lambda \geq 1$ such that:

$$\Lambda^{-1} \leq \frac{e(x, u)}{1 - e(x, u)} \bigg/ \frac{e(x)}{1 - e(x)} \leq \Lambda$$

The full odds of treatment can deviate from the observed odds by at most factor Λ .

Special cases

$\Lambda = 1$: $e(x, u) = e(x)$ for all u
 $\Rightarrow$ unconfoundedness holds.

$\Lambda > 1$: hidden confounding allowed.
Larger Λ = more confounding.

Relationship to Rosenbaum Γ

Γ bounds conditional odds between matched units. Λ bounds marginal odds relative to observed propensity. Both are odds-ratio models; Λ is more tractable for general regression-based estimation.

What does $\Lambda = 2$ mean? A concrete example

Suppose $e(x) = 0.5$ (observed odds = 1).

Under $\Lambda = 2$: full odds $\in [1/2, 2]$, so

$$e(x, u) \in [1/3, 2/3]$$

Interpretation: the hidden U can shift treatment probability from 33% to 67% for someone with observed $e(x) = 0.5$.

Under $\Lambda = 4$: $e(x, u) \in [0.2, 0.8]$ — very wide uncertainty.

Rule of thumb:

$\log \Lambda = 0.5 \Rightarrow \Lambda \approx 1.65$ (mild confounding)

$\log \Lambda = 1.0 \Rightarrow \Lambda \approx 2.72$ (moderate)

$\log \Lambda = 1.5 \Rightarrow \Lambda \approx 4.5$ (strong)

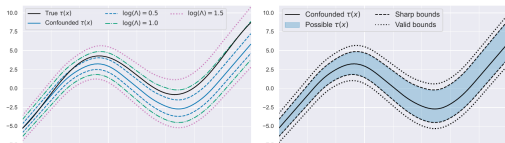
How to choose Λ ?

Two uses:

- 1 **Domain knowledge:** “How strong could the unmeasured confounder be?” E.g. for 401(k), could unobserved savings habit triple treatment odds?
- 2 **Breakdown analysis:** compute the Λ at which the policy conclusion (e.g. $\text{CATE} > 0$) first fails. Ask: is this Λ plausible?

Do **not** choose Λ after seeing the results to confirm a desired conclusion — that invalidates the analysis.

Valid vs. sharp bounds (Fig. 1)



Left: true $\tau(x)$ (black) stays inside sharp bounds at $\log \Lambda = 1.0$; outside at smaller Λ .

Right: sharp bounds (dashed) vs. valid bounds (dotted); blue = entire identified set.

Valid bound $\hat{\tau}^+(x)$

Covers $\tau(x)$ for all DGPs consistent with data and MSM. May include impossible values — overly cautious.

Sharp bound $\tau^+(x)$

The *tightest* valid bound.

Every value inside is achievable by some consistent DGP.

Sharp bounds describe the identified set exactly — no wasted space.

As $\Lambda \uparrow$: bounds widen.

At $\Lambda = 1$: bounds collapse to a point estimate (full identification).

The sharp CATE bounds (Result 1, from Dorn et al. 2021)

Set $\alpha = \Lambda/(\Lambda + 1) \in [0.5, 1)$. The sharp CATE upper bound is $\tau^+(x) = Y^+(x, 1) - Y^-(x, 0)$:

$$Y^+(x, 1) = e^*(x) \mu^*(x, 1) + (1 - e^*(x)) \rho_*^+(x, 1)$$

$$Y^-(x, 0) = (1 - e^*(x)) \mu^*(x, 0) + e^*(x) \rho_*^-(x, 0)$$

Identifiable nuisances:

- $e^*(x)$: propensity score
- $\mu^*(x, a) = \mathbb{E}[Y | X=x, T=a]$: outcome regression
- $\rho_*^\pm(x, a)$: **CVaR** terms (see next slide)

Lower bound

Swap $+ \leftrightarrow -$ and $1 \leftrightarrow 0$:

$$\tau^-(x) = Y^-(x, 1) - Y^+(x, 0)$$

Full interval: $[\tau^-(x), \tau^+(x)]$.

At $\Lambda = 1$: $\rho_*^\pm = \mu^*$, so $\tau^+(x) = \tau^-(x) = \tau(x)$.

Why CVaR? Hidden confounding as a distributional tilt

Question: what is the worst-case $\mathbb{E}[Y(1) \mid X = x]$ that the MSM allows?

Treated group ($T = 1$, fraction $e(x)$):

We *observe* their $Y(1)$.

The observed distribution $P(Y \mid X = x, T = 1)$ is **fixed by the data** for any $Q \in \mathcal{M}(\Lambda)$.

Untreated group ($T = 0$, fraction $1 - e(x)$):

$Y(1)$ is **never observed**.

U could have selected the units with *highest* $Y(1)$ into the control group.

Under MSM, U can tilt the selection by at most factor Λ — the worst case shifts mass to the **upper tail** of $Y(1)$.

The worst-case reweighting

The most extreme selection that Λ allows places maximum probability on the upper $(1 - \alpha)$ fraction of Y .

This is exactly:

$$\text{CVaR}_+(x, a) = \mathbb{E}[Y \mid Y \geq q_\alpha^*, X=x, T=a]$$

the **expected value above the α -quantile** (Conditional Value at Risk).

CVaR: formal definition and properties

Definition. $\text{CVaR}_+(x, a)$ at level $\alpha = \Lambda/(\Lambda + 1)$:

$$\text{CVaR}_+(x, a) := \mathbb{E}[H_+(Z, q_+^*) \mid X = x, T = a]$$

where the *CVaR pseudo-outcome* is:

$$H_+(z, \bar{q}) = \bar{q}(x, a) + \frac{\{y - \bar{q}(x, a)\}_+}{1 - \alpha}$$

and $q_+^*(x, a)$ is the true α -quantile of $Y \mid X = x, T = a$.

The ρ_*^+ in Result 1 combines both:

$$\rho_*^+(x, a) = \Lambda^{-1} \mu^*(x, a) + (1 - \Lambda^{-1}) \text{CVaR}_+(x, a)$$

Financial analogy

CVaR = “Expected Shortfall”: average loss in the worst α fraction.

Here: average $Y(1)$ in the tail that hidden U could force into the control group.

Effect of Λ

$\Lambda \uparrow \Rightarrow \alpha \uparrow \Rightarrow$ CVaR at more extreme quantile $\Rightarrow$ wider bounds.

$\Lambda = 1$: $(1 - \Lambda^{-1}) = 0$, so $\rho_{\pm}^* = \mu^*$
(CVaR *coefficient* is zero; the formula reduces without CVaR collapsing).

Sharp bound = outcome regression + CVaR correction

$$Y^+(x, 1) = \underbrace{e(x) \mu(x, 1)}_{\text{treated group: their } Y \text{ is observed fixed by data; no uncertainty here}} + \underbrace{(1 - e(x)) \rho^+(x, 1)}_{\text{untreated group: } Y(1) \text{ is counterfactual worst case = CVaR tail}}$$

Reading the formula:

- **Treated group** ($e(x)$ fraction): their $Y(1)$ is observed and matches $\mu(x, 1)$. The observed distribution is pinned by data, leaving no room for adversarial tilt.
- **Untreated group** ($1 - e(x)$ fraction): $Y(1)$ is missing. The MSM allows U to have preferentially placed the highest- $Y(1)$ individuals into the untreated group. Worst case = $\rho^+(x, 1)$ (CVaR blend).

At $\Lambda = 1$ (unconfounded)

$\rho^+(x, 1) = \mu(x, 1)$, so $Y^+(x, 1) = \mu(x, 1)$.
Bounds collapse: full point identification.

At large Λ

CVaR moves toward the supremum of Y .
Bounds widen: conclusion becomes uncertain.

Plug-in bias: why not just substitute estimated nuisances?

A natural approach: estimate $\hat{e}$, $\hat{\mu}$, $\hat{\rho}$, then plug into Result 1:

$$\hat{Y}^+(x, 1)_{\text{plug-in}} = \hat{e}(x) \hat{\mu}(x, 1) + (1 - \hat{e}(x)) \hat{\rho}_+(x, 1)$$

The problem. Suppose $\|\hat{e} - e^*\| = n^{-1/4}$ and $\|\hat{\rho}_+ - \rho_*^+\| = n^{-1/4}$ (slow but typical rates in semiparametric models). The plug-in bias is first-order:

$$\hat{Y}^+(x, 1)_{\text{plug-in}} - Y^+(x, 1) \sim |\hat{e} - e^*| + |\hat{\rho}_+ - \rho_*^+| = O(n^{-1/4}) \rightarrow 0 \text{ slowly}$$

Plug-in is biased for all n

The bias does not vanish at $\sqrt{n}$ rate. The estimator is **inconsistent** at $\sqrt{n}$ and cannot achieve confidence intervals without shrinking bias away.

The fix: Neyman-orthogonal (doubly-robust) pseudo-outcomes

Construct a corrected estimator whose bias is a **product** of nuisance errors: $|\hat{e} - e^*| \cdot |\hat{\rho}_+ - \rho_*^+|$. At $n^{-1/4}$ rates each, the product is $n^{-1/2}$ — fast enough for $\sqrt{n}$ -consistent estimation.

Neyman orthogonality and cross-fitting

Neyman-orthogonal score:

An estimating equation $m(\tau, \eta)$ is

Neyman-orthogonal if:

$$\partial_{\eta} m(\tau^*, \eta^*) [\hat{\eta} - \eta^*] = 0$$

The score is *first-order insensitive* to nuisance estimation errors.

Effect: the remaining bias is second-order (a product of rates), not first-order.

Cross-fitting (K-fold):

- 1 Split data into K folds.
- 2 Estimate nuisances on fold $-k$ (held-out).
- 3 Compute pseudo-outcomes on fold k only.

Decouples nuisance training from pseudo-outcome evaluation $\Rightarrow$ avoids overfitting

Quasi-oracle rate

When nuisances converge at $n^{-1/4}$:

Bias $\lesssim$ product $= n^{-1/4} \cdot n^{-1/4} = n^{-1/2}$

So the second-stage estimator converges at the rate it would if the nuisances were *exactly known* (oracle).

This is the **quasi-oracle property**.

Why $n^{-1/4}$?

Typical nonparametric rate in moderate dimensions. Both $\hat{\eta}$ and $\hat{\rho}_+$ must achieve this for the product-of-rates guarantee.

The B-Learner pseudo-outcome (Definition 2)

Applying the Kallus & Oprescu (2023) machinery to the Dorn et al. bound formula gives:

$$\begin{aligned}\phi_1^+(Z, \hat{\eta}) &= TY + (1 - T)\hat{\rho}_+(X, 1) \\ &\quad + \frac{(1 - \hat{e}(X))T}{\hat{e}(X)} (R_+(Z, \hat{q}_+) - \hat{\rho}_+(X, 1))\end{aligned}$$

$$\phi_{\tau}^+(Z, \hat{\eta}) = \phi_1^+(Z, \hat{\eta}) - \phi_0^-(Z, \hat{\eta})$$

What the IPW correction does

The term $\frac{(1 - \hat{e})T}{\hat{e}}(\cdot)$ reweights treated units to represent the untreated group's counterfactual. This is the first-order correction that makes the score Neyman-orthogonal.

Double robustness: validity

Bounds remain **valid** if either $\hat{e} \rightarrow e^*$ or $\hat{\rho}_+ \rightarrow \rho_+^*$ (one consistent nuisance suffices to cancel the product-of-rates bias).

Sharpness requires more

For **sharp** bounds, quantile estimates $\hat{q}_{\pm}$ must also be consistent. If $(\hat{q}_+ - q_+^*)^2 \not\rightarrow 0$, bounds remain valid but are no longer sharp — they become conservative.

At $\Lambda = 1$: reduces to the DR-Learner pseudo-outcome exactly.

B-Learner: the two-stage algorithm

Stage 1: nuisance estimation (K-fold cross-fitting)

- 1 Estimate propensity $\hat{e}(x)$
- 2 Estimate outcome regression $\hat{\mu}(x, a)$
- 3 Estimate quantile $\hat{q}_{\pm}(x, a)$ at level α
- 4 Construct $\hat{\rho}_{\pm}(x, a) = \Lambda^{-1}\hat{\mu} + (1 - \Lambda^{-1})\widehat{\text{CVaR}}_{\pm}$

Stage 2: bounds regression

- 5 Compute pseudo-outcomes $\hat{\phi}_{\tau}^{\pm}(Z_i, \hat{\eta}^{(k)})$ on each fold
- 6 Regress pseudo-outcomes on X using any learner $\mathbb{E}_n$
- 7 Output $\hat{\tau}^{+}(x)$ and $\hat{\tau}^{-}(x)$

Flexibility

Any learner for each stage: RF, NN, linear, kernel. Notation: $\hat{\tau}^{+}(\{1^{\text{st}}\}, \{2^{\text{nd}}\})$, e.g. $\hat{\tau}^{+}(\text{RF}, \text{RF})$.

Course connection

Stage 2 is identical to the DR-Learner, but with bounds pseudo-outcomes $\hat{\phi}_{\tau}^{\pm}$ instead of CATE pseudo-outcomes.

Extra cost vs. DR-Learner: quantile regression in Stage 1.

Theorem 1: the full bias bound

Theorem 1 (Pseudo-outcome Conditional Neyman Orthogonality)

Under Assumption 2 (boundedness), the absolute bias $|\mathcal{E}_\tau^+(x; \hat{\eta})|$ of the CATE upper-bound pseudo-outcome satisfies:

$$\begin{aligned} |\mathcal{E}_\tau^+| \lesssim & \underbrace{|\hat{e} - e^*| \cdot |\hat{\rho}_+(x, 1) - \rho_*^+(x, 1)|}_{\text{treatment side}} + \underbrace{(\hat{q}_+(x, 1) - q_*^+(x, 1))^2}_{\text{quantile, treatment}} \\ & + \underbrace{|\hat{e} - e^*| \cdot |\hat{\rho}_-(x, 0) - \rho_*^-(x, 0)|}_{\text{control side}} + \underbrace{(\hat{q}_-(x, 0) - q_*^-(x, 0))^2}_{\text{quantile, control}} \end{aligned}$$

Three terms, two mechanisms

Product terms: vanish if either $\hat{e}$ or $\hat{\rho}$ is consistent on the respective side.

Squared quantile terms: independent of $\hat{e}$ — must converge on their own.

Implication

Validity: one of $\hat{e}$ or $\hat{\rho}_+$ consistent $\Rightarrow$ product terms vanish.

Sharpness: also need $\hat{q}_\pm \rightarrow q_\pm^*$.
Applies independently to both sides.

Pointwise vs. average validity: an important subtlety

What Corollary 1 (ERM, in the paper) actually gives

For an ERM second-stage regression over a function class $\mathcal{F}$, validity holds **on average**:

$$\frac{1}{n} \sum_{i=1}^n [\hat{\tau}^+(X_i) - \tau^+(X_i)] \geq -o_p(1)$$

The estimated bound covers the true bound *in aggregate over the sample*, not necessarily at every individual x .

Pointwise validity requires more

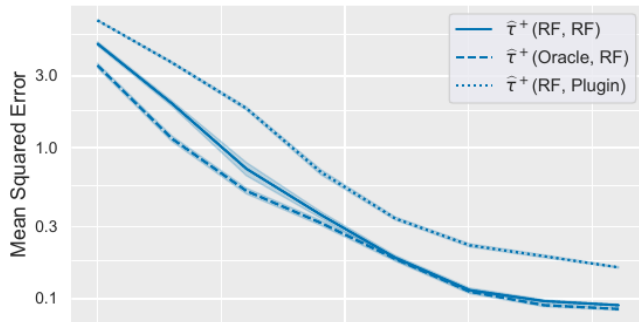
For $\hat{\tau}^+(x) \geq \tau^+(x) - o_p(1)$ at every x : Appendix C.2 shows this holds for **linear smoothers** only. General ERM (e.g. random forests) does not guarantee it.

What this means in practice

CATE bounds are most useful *per individual* — this requires pointwise validity.

Use linear smoothers in Stage 2 for pointwise guarantees. With ERM, interpret as average-valid only.

Empirical results: quasi-oracle convergence (Fig. 2a)

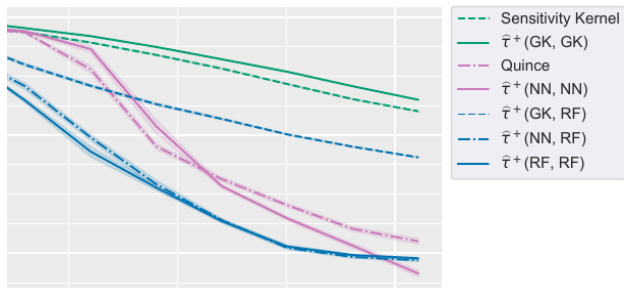


B-Learner (RF,RF) vs. oracle ($\hat{\tau}^+$ with true nuisances) vs. plug-in.

Key observation: B-Learner and oracle converge at the *same rate* (parallel lines, log-log scale). The plug-in is biased for every n .

This confirms the quasi-oracle property — with a few hundred samples, B-Learner already performs nearly as well as if it knew the propensity and CVaR exactly.

Empirical results: comparison to competitors (Fig. 2b)

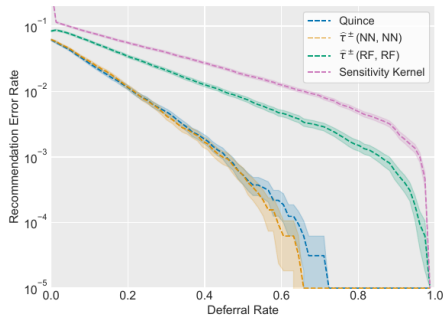


B-Learner vs. *Sensitivity Kernel* (Gaussian kernel nuisances) and *Quince* (Bayesian NN).

Using the same nuisances (GK), B-Learner is comparable to *Sensitivity Kernel*.

With NN or RF nuisances, B-Learner outperforms GK-based competitors — confirming that learner flexibility matters.

Empirical results: IHDP deferral policy (Fig. 3)



Setting: IHDP with hidden confounding.

Policy: recommend treatment if $[\hat{\tau}^-, \hat{\tau}^+]$ excludes zero; defer to expert otherwise.

x-axis: deferral rate (fraction deferred).

y-axis: recommendation error rate (log scale).

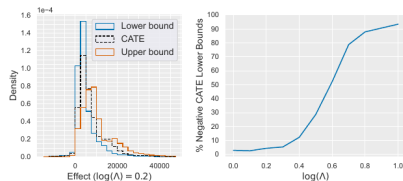
Result

B-Learner (NN,NN) matches *Quince*.

B-Learner (RF,RF) outperforms GK-based *Sensitivity Kernel*.

Mixing NN Stage 1 with RF Stage 2 also works well — a unique advantage of the meta-learner design.

Real data: 401(k) eligibility and wealth (Fig. 4)



Left ($\log \Lambda = 0.2$)

Distribution of lower/upper CATE bounds across individuals.

Most lower bounds positive $\Rightarrow$ likely real positive effect at low confounding.

How to read this figure

- **T:** 401(k) eligibility. **Y:** net financial assets.
- **Confounder:** unobserved saving preference.
- Left panel at $\log \Lambda = 0.2$: each person gets $[\tau^-(x), \tau^+(x)]$.
- $\tau^-(x) > 0 \Rightarrow$ effect **robust** at this confounding level.
- $\tau^-(x) \leq 0 \Rightarrow$ data do not rule out zero/negative effect.

Right (varying Λ) — breakdown analysis

As Λ grows, bounds widen; we track what fraction of lower bounds cross zero, asking how strong unobserved saving preference must be before the conclusion of a positive effect fails.

At $\log \Lambda \approx 0.5$: $\sim 50\%$ of lower bounds turn negative.
Breakdown question: is $\log \Lambda = 0.5$ plausible here?

Common mistake: “Confidence intervals already handle this”

What CI's do *not* capture

DR-Learner CIs quantify **statistical** uncertainty (finite-sample noise).

They assume unconfoundedness is true and *shrink* with n .

$\hat{\tau}(x) \pm 1.96 \text{ se}$ narrows as $n \rightarrow \infty$ — but converges to the *wrong* $\tau(x)$ if U exists.

What CATE bounds capture

$[\hat{\tau}^-(x), \hat{\tau}^+(x)]$ quantifies uncertainty from **hidden confounding**.

This uncertainty does *not* shrink with n if U is unresolvable (unidentifiability persists).

The two types of uncertainty are orthogonal — both are needed for trustworthy causal claims.

Second trap: pointwise vs. average validity

An ERM second stage gives average-valid bounds, not pointwise-valid ones. Do not interpret individual bounds as guaranteed to cover the true $\tau(x)$ pointwise without verifying the smoother conditions.

Part 2: Referee Mode

Reading the paper as a peer reviewer

Assumptions: tested vs. untested

Assumption	Status	Comment
MSM correctly models confounding	Untested	Observationally indistinguishable from alternatives
Positivity $e(x) \in (\epsilon, 1-\epsilon)$	Assumed	Standard; IPW unstable near boundaries
Consistency $Y = Y(T)$	Assumed	SUTVA; not discussed
Λ set correctly	User choice	No data-driven selection method
Nuisance boundedness	Assumed	Required for Neyman-orthogonality theory
Quantile accuracy	Partial	Needed for sharpness (not just validity)

The hardest assumption

The MSM, Rosenbaum Γ , e-value, and partial R^2 models all give *different* bounds under the same data. The choice of sensitivity model is itself untested and shapes all conclusions.

Limitations acknowledged and not acknowledged

Acknowledged:

- **Binary treatment only** ($T \in \{0, 1\}$): continuous and multi-valued T not covered.
- Λ is **user-specified**: too small $\Rightarrow$ invalid; too large $\Rightarrow$ vacuous.
- **Rate requires** $n^{-1/4}$ **nuisances**: slow in high-dimensional X .

Not addressed:

- **Pointwise validity**: ERM Stage 2 gives only average-valid bounds (see Appendix C.2 for linear-smoother case).
- **Simultaneous validity across all x** : no family-wise error control.
- **Sensitivity model selection**: no method for choosing between MSM and alternatives.
- **Computational cost**: quantile regression in Stage 1 is more expensive than mean regression.
- **Limited empirical improvement**: B-Learner only matches *Quince* in practice despite stronger theory.

Comparison to related methods

Method	Target	Properties	Flexibility
Rosenbaum sensitivity	ATE bound	Valid	Matched pairs
Sensitivity Kernel	CATE bound	Valid	GK only
Quince (Jesson 2021)	CATE bound	Valid	BNN only
B-Learner	CATE bound	Valid + sharp + quasi-oracle + doubly robust	Any learner

Where B-Learner wins: learner flexibility, theoretical guarantees (sharpness, quasi-oracle rate), both-sides bias bound.

Practical caveat: B-Learner(NN,NN) and *Quince* perform similarly in Fig. 3. Theoretical superiority does not always translate to empirical gains on small datasets.

Discussion questions

- 1 At $\log \Lambda = 0.6$, about half of 401(k) CATE lower bounds turn negative. **Does this mean the 401(k) effect is confounded?** What additional evidence would distinguish confounding from sampling variability?
- 2 The quasi-oracle property guarantees B-Learner converges at the oracle rate. **Does this mean B-Learner is safe to use on 200 observations?** What does the asymptotic constant hide?
- 3 Suppose you use bounds $[\hat{\tau}^-(x), \hat{\tau}^+(x)]$ to decide who gets a treatment. **How would you operationalize the decision?** What role does Λ play in the decision rule?
- 4 ERM in Stage 2 gives average-valid bounds, not pointwise-valid ones. **When does this matter?** Can you give a decision-making scenario where average validity is sufficient?
- 5 At $\Lambda = 1$, B-Learner reduces to DR-Learner. **What does this tell us about unconfoundedness?** Is it the same as saying the identified set is a single point?

Core contribution:

$$\hat{\tau}^+(x), \hat{\tau}^-(x) \approx \tau^+(x), \tau^-(x)$$

sharp CATE bounds under the MSM, at quasi-oracle rates, using any nuisance learner.

What it contributes

- First CATE sensitivity method: valid + sharp + efficient + doubly robust
- Lineage: Dorn et al. (ATE bounds) + Kallus & Oprescu (DML machinery) → CATE bounds
- At $\Lambda = 1$: reduces to DR-Learner
- Practical: bounds widen with Λ , narrow with n

What remains open

- Continuous and multi-valued treatments
- Data-driven selection of Λ
- Pointwise validity guarantees for ERM Stage 2
- Sharpness under alternative sensitivity models
- Behavior under covariate shift